# Kinwise.

Family medicine-review co-pilot · private, on-device AI

UnivaBio 2026 · AI for Human Health | Fictional demo data

**Video:** on the Devpost page · runs in the browser, no backend

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## Helps families caring for an older adult on many medicines notice what changes after a medicine change, and bring a clear question to the visit.

**Live demo:** albertzorzolan.github.io/UnivaBioBabson → Try as Alex  **Code:** github.com/albertzorzolan/UnivaBioBabson/tree/medwatch-v2

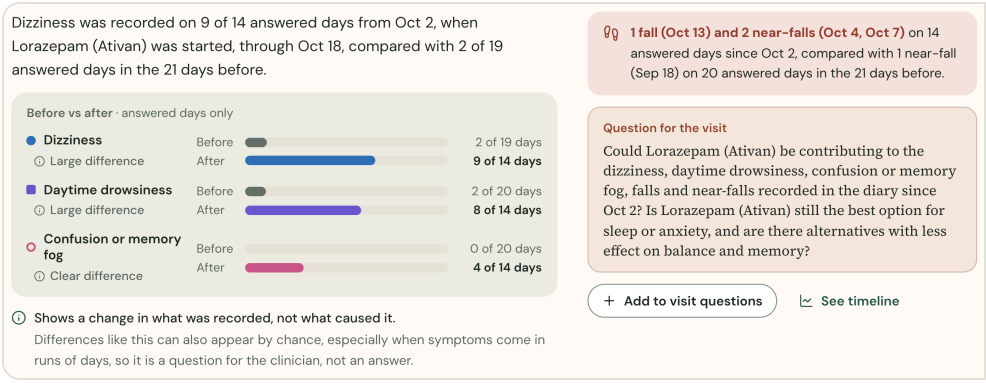
## The problem

**43.0%** NCHS, Health, United States
of US adults 65+ took **5 or more prescription drugs** in a 30-day period (2017–Mar 2020).

**37%** Gandhi et al., NEJM 2003
of adverse drug events in primary care whose harm could have been reduced were attributed to **the patient not telling the doctor**.

**More than 1 in 4** CDC, Facts About Older Adult Falls
older adults **fall each year**, but **less than half tell their doctor**.

**What's different.** Diary apps (MyTherapy, CareClinic) chart symptoms but do not document before/after counts around medicine starts; clinician tools (MedStopper, MedSafer) apply Beers/STOPP-type rules without the family's observations. Kinwise does both.[4]




### What Alex brings to the visit

The caregiver Overview in the latest build (fictional data; the live demo shows an earlier layout of this card): counts on answered days only, a reminder that a change is not a cause, and a question for Margaret's clinician.

## Who it is for, and where

The **family caregiver** (Alex) and the **older adult** (Margaret, 78, 8 medicines), who checks in on a large-text screen; the brief is for her **doctor or pharmacist**.

For the Medicare **Annual Wellness Visit** (whose cognitive check must consider concerns raised by family[1]) or a pharmacist's **medication review**, which may be done with the caregiver[2].

> **Why it could help:** in EMPOWER[3] (cluster RCT, 303 long-term benzodiazepine users aged 65–95), **62%** of those given a brochure raised stopping with their doctor or pharmacist, and **27% vs 5%** (brochure vs usual care) had discontinued at 6 months (the brochure included a taper plan; Kinwise leaves that to the clinician). A reason to test Kinwise, not proof it works.

## What it does

1. **Check in, in her own words.** On-device NLP turns "felt woozy" into answers she confirms.
2. **Review her medicines** against AGS Beers 2023, anticholinergic burden, fall-risk drugs and cascades; each finding is cited and ends as a question.
3. **See what changed.** Dizziness on **2 of 19** answered days before lorazepam, **9 of 14** after (Fisher's exact test). With no real change, about 1 in 10 simulated medicine changes still showed a large or clear difference.
4. **Bring one reviewed page to the visit,** plus a FHIR R4 export.

## How it works

Cited KB → Engines → LLM rewords → Code checks → Human review

Deterministic TypeScript engines over a cited knowledge base (427 medicines, 89 Beers rules, 26 cascades) compute every fact. An optional on-device LLM (Qwen2.5, WebGPU) only rewords them; code drops sentences that fail a number, name or wording check.

## Evidence & rigor

Each figure below can be reproduced from the code at medwatch-v2.

- **Clinical NLP:** micro-F1 **94.2%** on 70 held-out stories never used for tuning; dev set (245 stories) 99.8%.
- **Medication review: 57 of 57** synthetic cases pass (a written spec, not clinical validity), 295 of 295 expectations.
- **On-device AI:** 0 external requests in a recorded run; its review found flawed outputs that got through (ai-live.json), so AI stays optional.
- **Tests** (recorded 2026-10-11): 1,245 unit tests pass · 20 of 20 end-to-end checks pass · 46 axe scans, 0 violations.

### Working

- Public live demo, no install
- NLP check-in · medicine matching
- Cited review · before/after · Ask Kinwise
- Reviewed brief · print · FHIR R4 export

### Partial

- On-device LLM: optional, WebGPU only
- KB: 427 medicines, no clinician review
- No dose or allergy checks · English only
- One device, no sync · no FHIR receiver

### Planned (not built)

- Pharmacist KB review · validation pilot
- Encrypted family sharing and storage
- Pharmacy, portal import · voice · Spanish

Hackathon prototype: not FDA reviewed or cleared, not clinically validated, no HIPAA claim. Designed never to say a medicine caused a symptom or to advise a change; the clinician decides. Emergency: call 911.

[1] 42 CFR 410.15 (detecting cognitive impairment). [2] Part D Comprehensive Medication Review, when the patient cannot take part because of cognitive impairment: 42 CFR 423.153(d)(1)(vii)(B)(2). [3] Tannenbaum et al., JAMA Intern Med 2014. [4] Vendor pages, Oct 2026; "not documented" is not proof of absence.